

Sachin Bala

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Technical Skills

Python (Pandas, NumPy, scikit-learn), SQL (MySQL), R, Statistical Analysis, A/B Experimentation, DOE, SPC, Process Capability Studies, Data Visualization (Grafana, Matplotlib), Machine Learning, Computer Vision (YOLO), JMP, Minitab, LabVIEW, SolidWorks, Six Sigma, PFMEA, Lean Manufacturing, Microsoft Suite, Cognex, Dr Schenk vision system

Education

Western Michigan University (WMU)

Overall GPA: 3.69

Bachelor's in Mechanical Engineering with minor in Computer Science and Mathematics

Work Experiences

Associate Process Development Engineer, Tesla Inc., Austin, TX

02/25-present

- Commissioned two electrode production lines achieving 98% yield through systematic process experimentation and data-driven qualification protocols; defined and monitored real-time operational metrics to quantify performance trends and guide investment decisions.
- Applied quantitative analytical methods (DOE, SPC, FMEA, capability studies, 8D, 5-Why) to large-scale operational datasets; formulated hypotheses, designed experiments, and delivered actionable insights to engineering and business stakeholders.
- Engineered end-to-end analytics pipelines using MySQL, Python, R, and JMP to surface predictive insights into tool and process performance, directly enabling data-driven product and operational decisions across the manufacturing organization.

Engineering Technician, Process Development Team, Tesla Inc., Austin, TX

8/23-02/25

- Designed and deployed a real-time Grafana monitoring pipeline tracking tool performance metrics across the production floor, enabling proactive anomaly detection and reducing unplanned failures; authored SOPs for sensor integration and dashboard usage.
- Conducted statistical capability studies and A/B-style process experiments to validate tools against product specifications, driving a 10% increase in production throughput while sustaining quality targets.
- Performed structured root cause analysis on downtime events using SPC charts, fault logs, and process historians; quantified impact and implemented corrective actions that improved operational reliability and uptime.

Engineering Co-Op, Scherdel SST, Muskegon, MI

8/22-12/22

- Synthesized EV component market data and manufacturing material trends to support product launch decisions and quality process design, delivering quantitative findings to cross-functional stakeholders.

Process Engineering Intern, Scherdel SST, Muskegon, MI

05/22-08/22

- Built a predictive failure-detection model using ML algorithms and statistical analysis, improving manufacturing efficiency and accuracy by 25%; validated model using Gauge R&R methodology and F1-score metrics.
- Developed a Computer Vision defect detection system; evaluated model effectiveness using structured quantitative validation (Gauge R&R, F1-score analysis).

Laser Tech Technician Internship, Technique Inc., Jackson, MI

09/21-04/22

- Optimized laser cutting using CAD and machine troubleshooting; validated chassis parts via design review and laser tracker measurement data.
- Updated SOPs and trained technicians and operators to follow standardized procedures.

Engineering Research and Project

Senior Design Project - Two Tier Industrial Cart (Technique Inc.)

01/22-12/22

- Designed a two-tier industrial cart in SolidWorks with scissor lift mechanism; validated with FEA analysis and numerical calculations.
- Applied DFMEA to identify and mitigate risk, establishing data-backed design controls and a quality control plan.

Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List, Yellow Belt Training (Six Sigma), Root Nepal (Co-Founder)